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RESEARCH

Robust *Pichia kudriavzevii* for open production of methylotrophic protein

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Abstract

Methanol has emerged as a promising substrate in biomanufacturing, where high tolerance and efficient assimilation are critical to overcome the cytotoxic effects of methanol and its intermediate metabolites. In this study, a *Pichia kudriavzevii* P22 strain (genome size: 17.3 Mb) was isolated from fermented camel milk. Remarkably, this strain tolerated up to 15% (v/v) methanol and survived under multiple stress conditions, including pH 2, 47 °C, 5 g/L acetic acid, and 100 g/L NaCl. In a 5-L fed-batch fermentation system operated under open, non-sterile conditions with 7.5% (v/v) methanol as the sole carbon source, P22 remained viable for 78 h. The maximum cell dry weight (CDW) reached 27.56 g/L, with proteins accounting for 54%. Furthermore, a preliminary investigation of the methanol metabolic pathway in *P. kudriavzevii* P22 was performed. Collectively, this work identifies *P. kudriavzevii* P22 as a potential eukaryotic chassis for developing methylotrophic yeast cell factories. Importantly, it demonstrates an open, unsterilized methanol-based fermentation process at 5 L scale, offering a potential strategy for scale-up fermentation with simplified operation and reduced cost.

Key points

• *A P. kudriavzevii P22 strain capable of tolerating 15% (v/v) methanol was isolated from fermented camel milk.*

• *In an open, non-sterile 5-L fermentation system with methanol as the sole carbon source, P. kudriavzevii P22 achieved 27.56 g/L CDW with 54% protein content.*

Keywords Methylotrophic yeast · Single-cell protein · Open-unsterilized fermentation · Stress tolerance

Introduction

With the rapid growth of the global population and the increasing scarcity of resources, conventional protein production methods are facing severe challenges. Animal-based protein production requires large amounts of grain, land, and water, and contributes substantially to greenhouse gas emissions and environmental pollution during farming (Smith et al. 2024). Although plant-based proteins are relatively more resource-efficient, they still encounter limitations related to extraction efficiency, protein quality, and food security (Sá et al. 2020; Su et al. 2024). Besides, to address these issues, various efficient and sustainable protein alternatives have been developed. Among them, single-cell protein (SCP) has attracted considerable attention as an emerging protein source (Li et al. 2024). SCP contains all essential amino acids required by humans and can be produced through fermentation processes that efficiently convert diverse organic wastes and agricultural by-products

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(Salazar-López et al. 2022). Its production is environmentally friendly and energy-efficient, making it an attractive sustainable option for both food (Bajić et al. 2022) and animal feed (Vethathirri et al. 2021). Notably, microorganisms capable of utilizing one-carbon compounds in SCP production have shown great promise (Ye et al. 2024). One-carbon compounds such as methanol, methane, and carbon dioxide are abundant and often originate from industrial by-products or atmospheric greenhouse gases. These compounds can be efficiently converted into protein by specialized microorganisms, offering not only a novel approach for carbon emission mitigation but also a more economical and environmentally sustainable feedstock for protein production (Gao et al. 2025).

The production of single-cell protein (SCP) through fermentation relies on diverse microbial strains that convert carbon sources into protein-rich biomass (Sekoai et al. 2024). Microorganisms including yeasts, fungi, bacteria, and algae have been employed in SCP fermentation (Li et al. 2024; Salazar-López et al. 2022). Among these, yeasts are preferred because of their rapid growth, minimal nutritional requirements, and high environmental tolerance (Steensels et al. 2014). Different yeast strains exhibit distinct fermentation characteristics and adaptability, and in recent years, methanol fermentation has emerged as a particularly important and extensively studied field (Kelso et al. 2022; Vasudevan et al. 2025). Methanol, an inexpensive and abundant one-carbon compound, can be converted into high-quality SCP by yeast strains, thereby reducing production costs and improving the sustainability and environmental friendliness of the process (Meng et al. 2023). In addition, the high energy density and reductive potential of methanol render it an attractive fermentation substrate (Zang et al. 2025). Methylotrophic microorganisms, which utilize methanol as the sole carbon and energy source, offer a promising platform for future biorefining (Wegat et al. 2022). However, the development of methanol-based biomanufacturing is limited by the suboptimal genetic characteristics and restricted availability of genetic engineering tools for native methylotrophic microorganisms, which collectively hinder carbon conversion efficiency and product diversity (Sakarika et al. 2022). Moreover, methanol and its metabolic by-products are toxic to most microorganisms, for which many strains exhibit growth inhibition in methanol-containing media (Jia et al. 2024). A reduced metabolic activity under high methanol concentration is common even for methanol-utilizing strains, limiting their suitability for industrial-scale applications (Cotton et al. 2020; Fabarius et al. 2021). For example, though *Pichia pastoris* has been widely studied for its methanol metabolism capabilities and biosafety, its growth remains constrained under high-concentration methanol (Cai et al. 2022; Guo et al. 2023). Therefore, exploring novel strains with promising methanol utilization efficiency is essential, as it would enable high-value conversion of methanol and facilitate large-scale production across diverse fermentation models (Zhu et al. 2020).

The complexity of fermentation systems directly affects the production cost and operational efficiency in bioprocess design. In conventional closed fermentation systems, energy-intensive sterilization of equipment and raw materials is typically required to prevent microbial contamination (Wang et al. 2015). Such requirements not only increase energy consumption but also significantly elevate the production cost of bioproducts. By contrast, an unsterilized open fermentation system, which is characterized by lower cost, simplified operation, and reduced energy demand, has increasingly attracted research attention. In recent years, growing efforts have been devoted to developing sterilization-free open fermentation technologies and identifying microorganisms capable of thriving under extreme environments. For example, microorganisms tolerant to harsh conditions such as strong acids, strong bases, high salinity, and high temperature have emerged as promising candidates for open fermentation (Ye et al. 2023). Yue et al. reported a recombinant strain produced approximately 70% PHB over a 65-day open fermentation process (27 g/L NaCl, approximately pH 10, and 37 °C) without contamination (Yue et al. 2014). The methanol-tolerant strain *P. pastoris* has exhibited remarkable adaptability and growth performance, making it a suitable candidate for open, unsterilized fermentation (Meng et al. 2023).

In this study, a methanol-tolerant yeast strain was isolated from camel fermented milk in Xinjiang, which was identified as *P. kudriavzevii* P22 through systematic screening, identification, and morphological characterization. Transcriptomic analysis of P22 under high methanol concentrations was performed to investigate the mechanisms underlying exogenous protein (single-cell protein) expression. Additionally, fermentation conditions optimization was employed to establish the most suitable conditions for open, unsterilized fermentation, and the strain's capability to convert methanol into protein-rich biomass was explored. Overall, this study highlights the potential of the strains isolated from extreme environments in Xinjiang for open-unsterilized fermentation and proposes a novel methanol-based model for SCP production.

Materials and methods

Strain collection and purification

Nine traditionally fermented camel milk samples were collected from pastures around Fuhai County, Altay Prefecture, Xinjiang, and stored at 4 °C. All samples were processed within three days of collection to ensure microbial viability and data reliability. The isolation and purification of yeast were performed using the spread and streak plate method. Briefly, samples were inoculated into sterilized camel milk medium, activated by sequential cultivation, and serially diluted (10^{-2}–10^{-7}). Dilutions were plated, and after 48 h of incubation, 10–30 colonies were picked from each plate. Colonies were purified by streaking three times until morphology was uniform, and pure isolates were preserved on slants for subsequent experiments.

Culture medium and conditions

YPD medium, consisting of 20 g/L glucose, 20 g/L peptone, and 10 g/L yeast extract, was used for activation and purification in this study and was sterilized by autoclaving at 115 °C for 20 min. Meanwhile, WL Nutrient Agar medium (Qingdao Haibo Biotechnology Co., Ltd.) was supplemented with ampicillin sodium to a final concentration of 100 μg/mL and sterilized by autoclaving at 121 °C for 20 min. For dilution purposes, yeast suspensions were diluted with phosphate buffer (pH 7.2–7.4) containing magnesium sulfate, 0.001 g/L ferrous sulfate, 0.001 g/L zinc sulfate, and 0.001 g/L manganese sulfate. In addition, MSM medium comprised 5% methanol, 5 g/L ammonium sulfate, 0.1 g/L calcium chloride, 0.15 g/L potassium chloride, 0.2 g/L magnesium sulfate, 0.001 g/L ferrous sulfate, 0.001 g/L zinc sulfate, and 0.001 g/L manganese sulfate; it was sterilized by autoclaving at 121 °C for 20 min before use. For solid media preparation, 2% agar was added. In terms of culture conditions, plate culture was conducted at 30 °C; shake flask cultivation was performed at 30 °C with 200 rpm; and fermenter culture was conducted at 28 °C with 200 rpm.

Molecular biological identification of yeast

Yeast strains were streaked onto YPD solid medium and incubated overnight at 30 °C. Well-isolated colonies were transferred into 5 mL of YPD liquid medium and incubated overnight. 50 μL of the cultured medium was centrifuged at 12,000 rpm for 1 min, and the supernatant was discarded. The pellet was resuspended in a mixture of 90 μL of sterile ddH_2O and 10 μL of 1 M NaOH, vortexed vigorously, and incubated at room temperature for 10 min. Then, the pellet was collected by centrifugation (12,000 rpm, 1 min) and resuspended in 100 μL of sterile ddH_2O. The suspension was incubated in a 98 °C water bath for 10 min, then cooled to room temperature, diluted 10 times, and used for PCR amplification with 26S rDNA universal primers.

Agarose gel electrophoresis was employed to validate the PCR product size. The PCR products with the expected size were sent to Sangon Biotech (Shanghai) Co., Ltd. for sequencing. The resulting sequences were subjected to BLAST analysis in NCBI to retrieve 26S rDNA sequences of closely related species. A phylogenetic tree was then constructed using the Neighbor-Joining method in MEGA 7.0.

Optimization of *Pichia kudriavzevii* P22 culture conditions

P22 strain was streaked onto YPD solid medium and incubated for 18 h at 30 °C. A single colony was picked and transferred into 10 mL YPD liquid medium, cultured for 18 h in a shaker (30 °C, 200 rpm). The cultured medium was then transferred into 50 mL fresh YPD liquid medium at a ratio of 2% (v/v), incubated at different temperature (28,30, 37, and 42 °C) or pH (2, 3, 4, 5, 6, 7, 8, 9, 10) in shaker (200 rpm) for 48 h, each with 3 replicates. OD600 was then measured with a multi-function microplate reader and used for biomass evaluation. The *Pichia kudriavzevii* P22/3-1-20 used in this study is available from the corresponding author upon reasonable request.

Preparation of electron microscopy samples

P22 strains were inoculated into YPD liquid medium containing 7.5% (v/v) methanol and cultured until the logarithmic growth phase. A 25 mL incubated aliquot was transferred to a 50 mL centrifuge tube and centrifuged at 5000 rpm for 5 min. Cells were collected by centrifugation (5000 rpm, 5 min), washed twice with 50 mL sterilized water. Subsequently, 2.5% glutaraldehyde fixative (40 times the volume of the yeast cells) was added, fixed overnight at 4 °C. A portion of the sample was then sent to the Science Compass Testing Platform for transmission electron microscopy (TEM). The remaining cells were collected by centrifugation, washed three times with 20 mL 1 × PBS buffer. Gradient dehydration was subsequently exploited with 30%, 50%, 70%, 80%, 90%, and 100% absolute ethanol sequentially, with each dehydration step lasting 10 min. After dehydration, the samples were subjected to substitution with an ethanol-tert-butanol solution (1:1, v/v) for 20 min. The substituted samples were freeze-dried for 48 h, then mounted on silicon wafers, sputter-coated with gold using an ion sputter coater, and finally observed via scanning electron microscopy.

Screening of methylotrophic yeasts

The screened yeast strains were individually inoculated onto MSM solid media with 0%, 5%, and 10% (v/v) methanol as the sole carbon source (with the control group using MSM solid medium containing 2% glucose as the sole carbon source). The strains were cultured at 30 °C for 7 days, and the survival and growth performance were observed and recorded. For further characterization, the strains were inoculated into MSM liquid media with methanol as the sole carbon source and cultured at 30 °C for 7 days. Every 24 h, 200 µL of the yeast suspension was transferred to a 96-well plate to determine the OD_{600} value, thus characterizing the growth kinetic characteristics of the strains in the methanol-containing liquid media.

Substrate utilization analysis

Activated strains were inoculated into MSM liquid media containing different carbon sources (glucose, fructose, raffinose, glycerol, starch, xylose, sucrose, lignocellulose, and xylan, respectively), incubated in a constant-temperature shaker at 30 °C and 200 rpm for 48 h, measuring OD_{600}. Subsequently, the data were analyzed to evaluate the metabolic utilization capabilities of strain P22 for different carbon sources.

Transcriptome analysis

Yeast was cultured in YNB medium with either 20 g/L glucose (control) or 7.5% (v/v) methanol (experimental). Cells in the logarithmic phase were collected, washed twice with sterile water, and stored at −80 °C. Total RNA was extracted using TRIzol reagent (Invitrogen), and samples with RIN (RNA Integrity Number) > 6.5 (Agilent 2100 Bioanalyzer) were used for library construction. cDNA libraries were prepared by Majorbio (Shanghai, China) and sequenced on the Illumina HiSeq 2000 platform. Gene expression was quantified and analyzed using RSEM (https://deweylab.github.io/RSEM/), normalized to TPM (Transcripts Per Million), and differential expression was analyzed with DEGseq (https://bioinfo.au.tsinghua.edu.cn/software/DEGseq), with $|\log_2 FC| \geq 1$ and $P < 0.05$ as the cutoff.

Determination of protein content and analysis of amino acid composition

Cells were collected during the exponential phase of fermentation in YPD medium supplemented with methanol. A 25 mL aliquot of the yeast suspension was transferred to 50 mL centrifuge bottles in batches and centrifuged at 5000 rpm for 5 min, discarded the supernatant, and washed twice with sterilized water. The collected yeast cells were freeze-dried overnight using a vacuum freeze dryer (FD-1A-50, Boyikang Instruments Co., Ltd., Beijing). Exactly 0.1 g of the dried cells was weighed for protein content determination using an automatic Kjeldahl nitrogen analyzer (K1160+K1124, Haineng Instruments Co., Ltd., Shandong). Additionally, for amino acid analysis, 30 mg of the solid sample was accurately weighed, and the amino acid composition and content of yeast proteins were analyzed using an automatic amino acid analyzer (Biochrom 30+, Biochrom Ltd., UK).

Fermentation system and method for open-unsterilized fermentation

In shake flask experiments, purified strains were activated to an OD_{600} value of 1 and inoculated into 250 mL Erlenmeyer flasks (with a working volume of 1/5) at a 2% (v/v) inoculum size. Cultures were incubated at 200 rpm for 48 h under different temperature conditions (26, 28, 30, 37, 45, 47, and 50 °C) to determine the optimal fermentation temperature. Similarly, purified strains were activated to an OD_{600} value of 1 and inoculated into 250 mL Erlenmeyer flasks (working volume 1/5) at a 2% (v/v) inoculum size. The pH was adjusted to values of 2, 3, 4, 5, and 6, and cultivation was performed at 28 °C and 200 rpm for 48 h to determine the optimal fermentation pH. For open shake flask fermentation, purified strains ($OD_{600} = 1$) were inoculated into 250 mL Erlenmeyer flasks (working volume 1/5) at a 2% (v/v) inoculum size, with YPD media containing 5%, 7.5%, and 10% (v/v) methanol, respectively. Fermentation was conducted at 28 °C and 200 rpm, and samples were taken every 24 h to determine the OD_{600} value and cell dry weight (CDW). Furthermore, purified strains ($OD_{600} = 1$) were inoculated into a 5 L fermenter (working volume 3/5) at a 2% (v/v) inoculum size, with YPD medium containing 7.5% (v/v) methanol. Fermentation was carried out at 28 °C and 200 rpm, and samples were taken every 2 h to measure the OD_{600} value, with sampling continued for 72 h. In the fed-batch fermentation process, the methanol content in the fermentation broth was determined by gas chromatography, and the amount of methanol to be supplemented was determined through preliminary experiments. Stress conditions were induced by adding 7.5% (v/v) methanol to the normal medium. When the strain reached the stationary phase at 32 h, methanol was supplemented to maintain the methanol concentration in the fermentation system at 5%. During this period, samples were taken every 2 h to measure the OD_{600} value. Microbial detection of the samples was performed using the streak plate method to detect foreign bacterial contamination. Meanwhile, PCR was used to amplify the conserved ITS and 16S regions that might be present in the samples, thereby ensuring the purity and reliability of the fermentation process. Open fed-batch fermentation was continued for 78 h.

RT-qPCR validation of *Pichia kudriavzevii* P22

One milliliter of Trizol was added to each centrifuge tube, and the mixture was vortexed thoroughly to lyse the cells, followed by the addition of 200 µL of chloroform and further mixing. After standing for 10 min, the mixture was centrifuged at 12,000 rpm at 4 °C for 10 min. A 400 µL aliquot of the supernatant was transferred to a clean, enzyme-free, and sterile 1.5 mL centrifuge tube, and 0.6 mL of isopropanol was added and mixed. After standing for 10 min, the mixture was centrifuged at 12,000 rpm at 4 °C for 10 min to remove the supernatant, yielding an RNA precipitate. The precipitate was washed with 1 mL of 75% ethanol solution (prepared with absolute ethanol and DEPC-treated water), followed by centrifugation at 12,000 rpm at 4 °C for 5 min to remove the supernatant; this washing step was repeated once. The resulting RNA precipitate was dried at room temperature for 10 min and then resuspended in 20–40 μL of DEPC-treated water to obtain an RNA solution. All steps, except the final rehydration step, were performed on ice, and all reagents were pre-cooled in advance. Finally, cDNA was synthesized using a reverse transcription kit.

Table 1 Growth of yeast strains in MSM solid medium with methanol as the sole carbon source

Strain number	5% methanol	10% methanol
2-1-5	+	-
2-1-10	+	-
2-1-13	+	-
2-2-14	+	-
2–8-15	+	+
3-1-9	+	-
3-1-18	+	-
3-1-20	+	+
3-2-7	+	-
3-2-8	+	-
3-3-7	+	-
3–4-3	+	+
3–4-4	+	-
3–4-6	+	+
3–4-15	+	-

"+" indicates that it can grow; "-" indicates that it cannot grow

Results

Screening of methylotrophic yeast

Fermented camel milk was sampled from the Altay region. A total of 134 yeast strains were isolated and purified from these samples using WL medium. To identify methylotrophic yeast, MSM medium—free of yeast extract, peptone, and other components that might interfere with screening results—was chosen as the screening medium for methylotrophic yeasts. 15 strains (Table 1) that could grow with 5% (v/v) methanol as the sole carbon source were identified when screened on MSM solid media supplemented with gradient methanol concentrations (0, 5, and 10% (v/v)). And 4 of them (2–8-15, 3-1-20, 3–4-3, and 3–4-6) were capable of retrieving normal growth on MSM solid medium with 10% (v/v) methanol. Subsequently, further characterization of these 4 strains was conducted using MSM liquid medium containing 5% (v/v) methanol as the sole carbon source. As shown in Fig. 1a, strains 3-1-20 and 3–4-6 exhibited superior growth characteristics compared to strains 3–4-3 and 2–8-15. Notably, after 120 h of cultivation, strain 3-1-20 accumulated a higher biomass ($OD_{600}=0.138$), which was approximately 1.6-fold that of strain 2–8-15 ($OD_{600}=0.085$).

P. pastoris GS115 is a commonly used industrial yeast strain. To explore the application potential of the screened strains, a growth capability evaluation was executed among strains 3-1-20, 3-4-6, and GS115 in MSM liquid medium containing 5% methanol, supplemented with 1% yeast extract and 2% peptone. As shown in Fig. 1b, all three yeast strains remained in the exponential growth phase within the first 36 h and reached the stationary phase at 36 h. Consistently, the growth performance of strains 3-1-20 and 3–4-6 was superior to that of *P. pastoris* GS115, Among them, the OD_{600} value of strain 3-1-20 is approximately 1.135 times

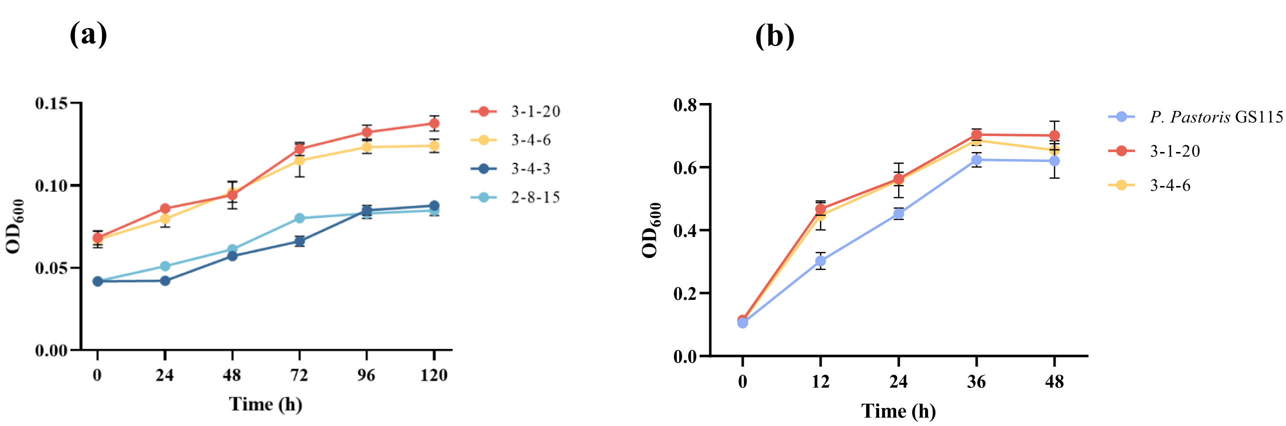

Fig. 1 **a** Growth of cultivable yeast strains in MSM liquid medium with 5% (v/v) methanol as the sole carbon source. **b** Growth of strain 3-1-20 in MSM medium with 5% methanol as the sole carbon source with 1% yeast extract and 2% peptone added

that of GS115. As a result, strain 3-1-20 was selected for subsequent studies.

Identification and morphological structure observation of methyl-trophic yeast

Molecular biological identification was used to identify the exact species of the screened strain 3-1-20. The ITS region and the D1/D2 region of 26S rDNA were PCR amplified from genomic DNA. The agarose gel electrophoresis results indicated that both PCR products had a length of approximately 500 bp (Fig. SF1), which was consistent with previous research. A further sequencing result revealed that it shared the highest sequence homology with *P. kudriavzevii* NRRL Y-5396, with a similarity of 99.47%. Furthermore, phylogenetic tree analysis based on the 26S rDNA D1/D2 region sequence (Fig. 2a) validated that strain 3-1-20 was closely related to *P. kudriavzevii* in terms of evolutionary relationship. Therefore, this yeast strain was identified as *P. kudriavzevii* and designated as *P. kudriavzevii* P22.

To investigate the morphology, structure, and function of *P. kudriavzevii* P22 at different perspectives and resolution levels, plate observation, scanning electron microscopy (SEM), and transmission electron microscopy (TEM) were conducted. After 48 h of cultivation on solid medium, the colonies of *P. kudriavzevii* P22 appeared white, with a flour-like, dry surface, smooth edges, and pseudohyphae (Fig. 2b). Subsequently, SEM was used to observe the surface morphology of *P. kudriavzevii* P22. As shown in Fig. 2c and d, the cell surface was clearly distinguishable at a magnification of 7000×. After gold sputtering, the sample surface exhibited good conductivity, and no significant deformation artifacts caused by dehydration-induced collapse or charging effects were observed, thus indicating that the pretreatment effectively preserved the original cellular morphology. Under methanol conditions, the cells displayed a distinct rod-like shape (Fig. 2d), in contrast to the ovoid cells observed under normal culture conditions (Fig. 2c). Next, TEM was employed to examine the internal structure of *P. kudriavzevii* P22. As shown in Fig. 2e and f, after cell fixation, embedding, sectioning, and staining, the organelle

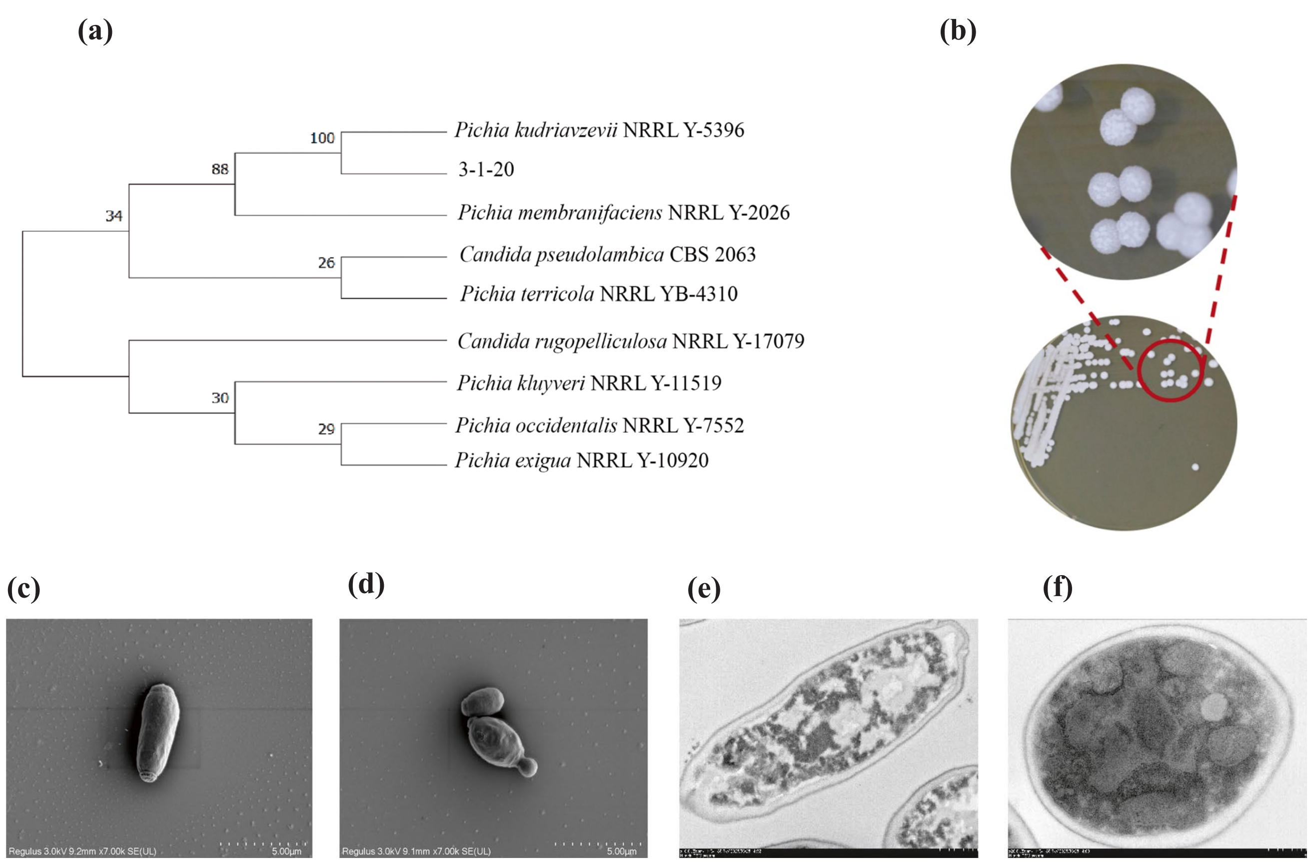

Fig. 2 **a** Phylogenetic tree of 26S rDNA D1/D2 sequences of strain 3-1-20. **b** The growth of yeast strain *P. kudriavzevii* P22 on a solid medium with 2% glucose as the carbon source. **c** SEM image of *P. kudriavzevii* P22 cultured in 2% glucose medium. **d** SEM image of *P. kudriavzevii* P22 cultured under 7.5% (v/v) methanol conditions. **e** TEM image of *P. kudriavzevii* P22 cultured in 2% glucose medium. **f** TEM image of *P. kudriavzevii* P22 cultured under 7.5% (v/v) methanol conditions

membrane structures showed significant contrast, and the intracellular ultrastructures were well preserved, with no obvious section folds or staining precipitate artifacts. Under 7.5% (v/v) methanol conditions (Fig. 2e), the internal structure of *P. kudriavzevii* P22 cells differed markedly from that under normal conditions (Fig. 2f). Specifically, under methanol stress, the intracellular structures were tightly aggregated with large cavities, whereas under normal conditions, the intracellular components were relatively evenly distributed.

Determination of tolerance for *Pichia kudriavzevii* P22

P. kudriavzevii is widely distributed in nature, exhibiting not only excellent acid and high-temperature tolerance but also maintaining superior fermentation properties under multiple stress conditions. To this end, to investigate the methanol tolerance of the screened *P. kudriavzevii* P22 strain in this study, the industrial strain *P. pastoris* GS115 was used as a control to evaluate the maximum threshold of methanol tolerance of *P. kudriavzevii* P22. The results showed that *P. kudriavzevii* P22 could grow on MSM solid medium with 15% (v/v) methanol as the sole carbon source, indicating that its methanol tolerance concentration reached up to 15% (v/v). In contrast, *P. pastoris* GS115 failed to grow on MSM medium with 15% (v/v) methanol as the carbon source. Collectively, these results confirmed that the methanol tolerance of *P. kudriavzevii* P22 was significantly superior to that of *P. pastoris* GS115 (Fig. 3a–d). This finding provides a valuable strain resource for subsequent studies on methanol tolerance-related molecular mechanisms and the construction of methylotrophic yeast cell factories.

Acetic acid, as one of the main by-products of yeast cells in biological manufacturing processes, can inhibit yeast growth and final product yield. Therefore, investigating the acetic acid tolerance of yeast strains is of great significance. Herein, the standard strain *Saccharomyces cerevisiae* S288C was included in the acid tolerance test, along with *P. pastoris* GS115 and *P. kudriavzevii* P22. Figure 3e–h show that all three yeast strains could tolerate 1 g/L acetic acid. Specifically, among them, *P. pastoris* GS115 exhibited the weakest acetic acid tolerance; *S. cerevisiae* S288C could tolerate 3 g/L acetic acid; and *P. kudriavzevii* P22 showed the strongest acetic acid tolerance, as it could still grow under 5 g/L acetic acid conditions.

During fermentation, high concentrations of salt ions in the fermentation system can reduce contamination by miscellaneous bacteria, enabling non-sterilized open-unsterilized fermentation and thereby mitigating cost increases caused by miscellaneous bacterial contamination. For this reason, the sodium chloride tolerance of the screened *P. kudriavzevii* P22 was tested using gradient concentrations of sodium chloride as one of the substrates in the medium. As shown in Fig. 3i–l, *S. cerevisiae* S288C, *P. pastoris* GS115, and *P. kudriavzevii* P22 could all survive under 10 g/L sodium chloride conditions. Furthermore, under 50 g/L sodium chloride conditions, both *P. kudriavzevii* P22 and *S. cerevisiae* S288C could grow, but the growth of *S. cerevisiae* S288C was slightly impaired. Notably, *P. kudriavzevii* P22 exhibited the optimal sodium chloride tolerance, as it could still grow under 100 g/L sodium chloride conditions, which

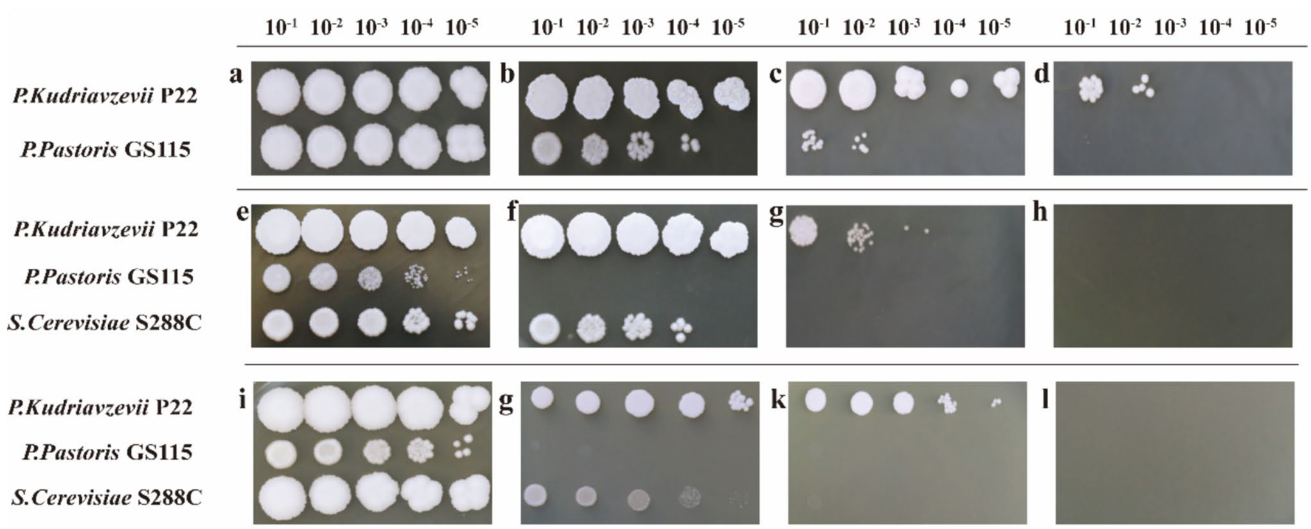

Fig. 3 **a–d** Tolerance of *P. kudriavzevii* P22 and *P. pastoris* GS115 to methanol (0%, 5%, 10%, 15%). **e**, **f** Tolerance test of *P. kudriavzevii* P22, *P. pastoris* GS115, and *S. cerevisiae* S288C yeasts to acetic acid (1 g/L, 3 g/L, 5 g/L, 10 g/L). **i–l** Tolerance test of *P. kudriavzevii* P22, *P. pastoris* GS115, and *S. cerevisiae* S288C yeasts to NaCl (10 g/L, 50 g/L, 100 g/L, 120 g/L)

provides a potential chassis cell for the subsequent construction of open non-sterilized fermentation systems based on yeast cell chassis.

Protein content determination and amino acid analysis of *Pichia kudriavzevii* P22

Protein content is a key indicator for evaluating high-quality single-cell proteins. The cellular protein content of *P. kudriavzevii* P22, determined via the Kjeldahl method, was 54%. Additionally, the amino acid composition and content of *P. kudriavzevii* P22 cellular proteins were analyzed (Table 2), with a total of 17 amino acids quantified. These include 7 essential amino acids (EAAs) required by the human body (lysine, phenylalanine, methionine, threonine, isoleucine, leucine, valine) and 10 non-essential amino acids (NEAAs) (tyrosine, histidine, aspartic acid, serine, glutamic acid, proline, glycine, alanine, cystine, arginine). The total amino acid (TAA) content reached 494.799 mg/g, which was significantly higher than the 360 mg/g protein specified in the FAO/WHO amino acid scoring standard and slightly higher than the 490 mg/g protein in the whole egg protein amino acid scoring standard. Specifically, among these, total non-essential amino acids (178.48 mg/g) accounted for 36.07% of the total amino acids. Among all measured amino acids, glutamic acid (Glu) had the highest proportion, accounting for 19.52% of total amino acids at 91.636 mg/g, followed sequentially by aspartic acid (Asp, 48.761 mg/g) and the essential amino acid lysine (Lys, 42.879 mg/g). In contrast, cystine (6.529 mg/g) and methionine (7.314 mg/g) were present in relatively low amounts.

Table 2 Amino acid components and Contents in *P. kudriavzevii* P22 cell protein

Amino acid types	Content (mg/g)	Proportion (%)
Glu	91.636	18.520
Asp	48.761	9.855
Lys	42.879	8.666
Leu	34.095	6.891
Arg	31.89	6.445
Ala	30.721	6.209
Ser	26.808	5.418
Thr	25.774	5.209
Val	23.823	4.815
Gly	23.77	4.804
Phe	22.883	4.625
Ile	21.712	4.388
Tyr	20.993	4.243
Pro	20.672	4.178
His	14.539	2.938
Met	7.314	1.478
Cys	6.529	1.320
EAA	178.48	36.071
NEAA	316.319	63.929
EAA/NEAA		56.424
TAA	494.798	

Transcriptome analysis of *Pichia kudriavzevii* P22

Analysis of differential gene expression in *P. kudriavzevii* P22 cultured with methanol versus glucose as the sole carbon source revealed via a volcano plot (Fig. 4a) a total of 685 significantly upregulated genes, 606 significantly downregulated genes, and 3615 genes with no significant differences. Notably, among the significantly upregulated genes, several were associated with amino acid synthesis and ATP synthesis/transport. Furthermore, GO secondary functional annotation analysis elucidated the adaptive mechanisms of *P. kudriavzevii* P22 in a methanol environment and its potential application value. As shown in Fig. 4b, *P. kudriavzevii* P22 exhibited significant differential gene expression across multiple functional categories, particularly in the following key categories. In ribosome biogenesis and RNA processing, genes related to ribosome biogenesis, rRNA metabolic processes, and RNA processing were significantly enriched, indicating that *P. kudriavzevii* P22 may adapt to environmental stress under methanol conditions by enhancing ribosome biogenesis and RNA processing capabilities. The upregulation of these genes may help maintain efficient protein synthesis, supporting cellular growth and metabolic demands in a methanol environment. Additionally, enrichment of genes related to ribonucleoprotein complex biogenesis further suggests that yeasts may require enhanced assembly and function of ribosomes and other RNA-protein complexes under methanol conditions to cope with stress from metabolic reprogramming. In parallel, in oxidoreductase activity and respiratory chain complexes, genes associated with oxidoreductase activity and respiratory chain complexes were significantly enriched, indicating that *P. kudriavzevii* P22 may adapt to methanol metabolism by enhancing oxidoreductase activity and respiratory chain function under methanol conditions. The upregulation of these genes may promote methanol oxidation and further metabolism of formaldehyde, thereby providing energy and intermediate metabolites for cells. *P. kudriavzevii* P22 achieved metabolic network reconfiguration and enhanced stress responses by upregulating genes related to ribosome biogenesis, RNA processing, oxidoreductase activity, and transmembrane transport complexes. This metabolic reprogramming enables cells to efficiently utilize methanol as a carbon source while enhancing tolerance to methanol-induced environmental stress. Similarly, KEGG enrichment analysis classified genes by their involvement in metabolic and biological processes, reflecting the metabolic networks

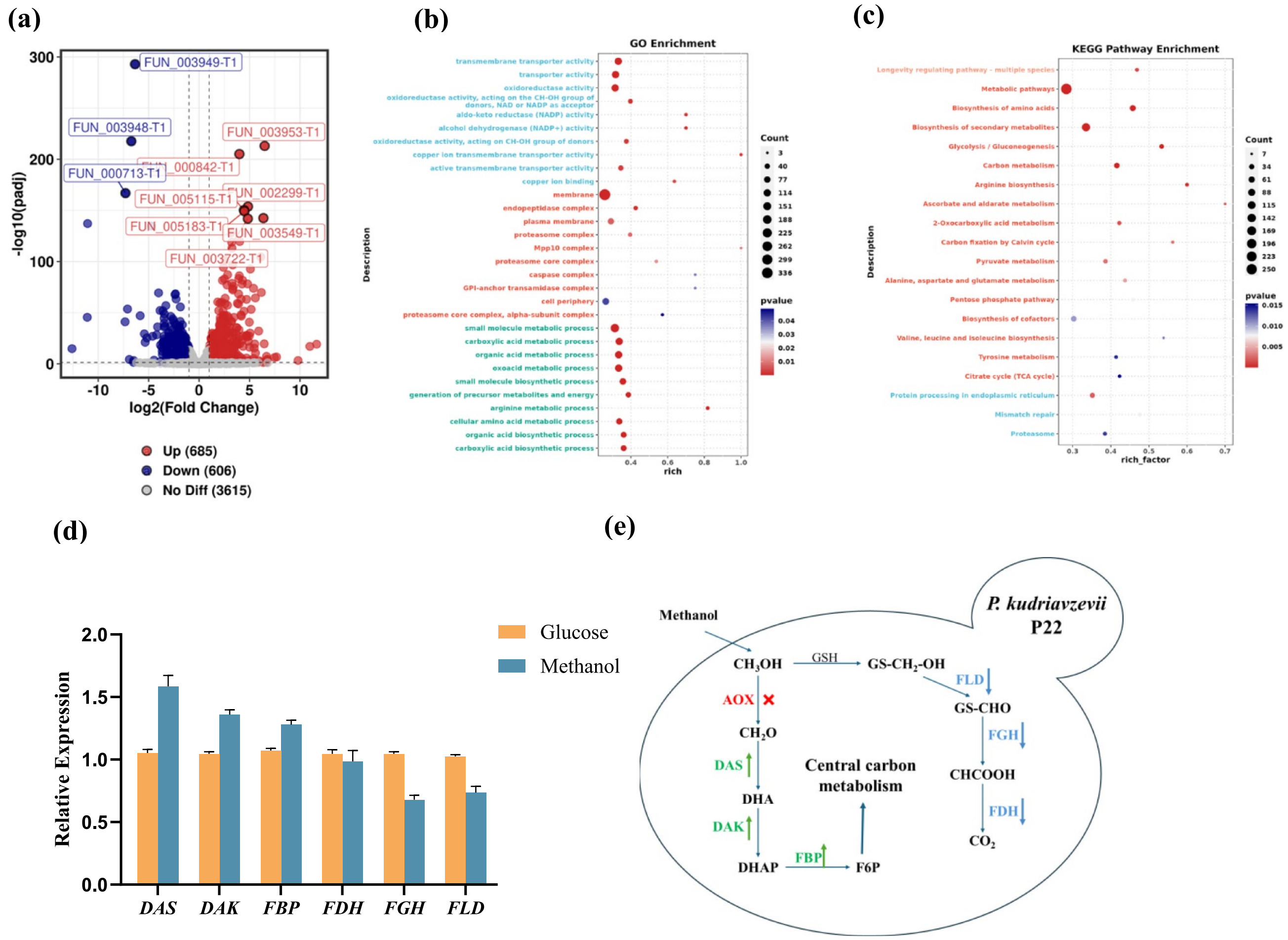

Fig. 4 **a** Differential gene volcano diagram. **b** GO enrichment analysis bubble chart. **c** KEGG pathway enrichment analysis Bubble chart. **d** RT-qPCR was used to verify the expression of the key gene of *P. kudriavzevii* P22 under methanol conditions. **e** The possible methanol metabolism pathway in *P. kudriavzevii* P22

and functional regulation of *P. kudriavzevii* P22 under different conditions. As shown in Fig. 4c, *P. kudriavzevii* P22 exhibited significant differential gene expression across multiple metabolic and biological processes, particularly in the following key pathways: Metabolic pathways, including enrichment of carbon metabolism, pyruvate metabolism, and glycolysis, reflect that *P. kudriavzevii* P22 exhibits intensive metabolic activity under methanol conditions to maintain its growth demands. Biosynthesis of amino acids and secondary products indicates high amino acid synthesis intensity in *P. kudriavzevii* P22 under methanol conditions, especially for arginine; meanwhile, *P. kudriavzevii* P22 may produce specific secondary metabolites to cope with environmental stress or enhance adaptability. Overall, *P. kudriavzevii* P22 exhibits significant metabolic reprogramming and stress response mechanisms under methanol as the sole carbon source, achieving efficient methanol utilization and environmental adaptation through regulation of multiple molecular and cellular processes.

Four natural methanol metabolic pathways are currently known: the serine pathway, ribulose monophosphate pathway (RuMP pathway), ribulose bisphosphate pathway (RuBP pathway), and xylulose monophosphate pathway (XuMP pathway). Methylotrophic bacteria utilizing methanol as a carbon source primarily employ the serine pathway, RuMP pathway, and RuBP pathway for methanol utilization, while methylotrophic fungi such as yeasts utilize the XuMP pathway. In this study, the XuMP pathway was first used as a template to analyze the methanol metabolic pathway in *P. kudriavzevii* P22. Key enzymes in the XuMP pathway—including alcohol oxidase (AOX), dihydroxyacetone synthase (DAS), dihydroxyacetone kinase (DAK), and fructose bisphosphatase (FBP)—were analyzed. These enzymes were first identified via alignment with *P. kudriavzevii* P22 genome data. Additionally, key enzymes potentially involved in yeast catabolic pathways—formaldehyde dehydrogenase (FLD), S-formylglutathione hydrolase (FGH), and formate dehydrogenase (FDH)—were searched and analyzed.

Results showed that the genome of *P. kudriavzevii* P22 annotated all the above key enzymes. Subsequent analysis of their expression levels revealed that no expression of the gene encoding AOX (the first key enzyme in the XuMP pathway) was detected in transcriptome data, while the expression levels of genes encoding key enzymes in methanol assimilation (DAS, DAK, FBP, etc.) were all upregulated. In contrast, the expression levels of genes encoding key enzymes in the methanol catabolic pathway (FLD, FGH, FDH) were downregulated. Subsequently, this study analyzed the other three natural methanol metabolic pathways. First, key enzymes in the serine pathway include serine hydroxymethyltransferase (SHMT), phosphoserine aminotransferase (PSAT), hydroxypyruvate reductase (HPR), glycerate kinase, phosphoenolpyruvate carboxylase (PEPC), and malyl-CoA lyase. Genes encoding SHMT, PSAT, HPR, and glycerate kinase were annotated in the genome, though their expression levels were downregulated. Next, the RuMP pathway was analyzed, focusing on its key enzymes: 3-hexulose 6-phosphate synthase (HPS) and 6-phospho-3-hexuloisomerase (PHI). Genes encoding HPS and PHI were annotated in the genome, but transcriptome data showed significantly decreased expression of these genes. Finally, key enzymes in the RuBP pathway—methanol dehydrogenase (MDH), formaldehyde dehydrogenase (FALDH), formate dehydrogenase (FDH), ribulose-1,5-bisphosphate carboxylase/oxygenase (Rubisco), and phosphoribulo kinase (PRK)—were analyzed. Except for FDH, none of these enzymes were annotated in the genome. Transcriptome analysis indicated decreased FDH expression.

To experimentally validate these transcriptome-based findings, this study further employed real-time quantitative polymerase chain reaction (RT-qPCR) for targeted validation of the expression levels of genes encoding key enzymes in this metabolic pathway. Two culture conditions were set: glucose as the sole carbon source (Glucose group) and methanol as the sole carbon source (Methanol group). By comparing differential expression of key genes between the two groups, the gene regulatory characteristics of the strain during methanol metabolism were revealed, with relevant results shown in Fig. 4d. RT-qPCR results showed that among key genes in the methanol metabolic pathway, no AOX expression signal was detected under either carbon source condition, suggesting that this gene may not participate in core regulation or is silenced during methanol metabolism in *P. kudriavzevii* P22. In contrast, compared with the Glucose group, the expression levels of DAS, DAK, and FBP in the Methanol group were significantly upregulated under methanol as the sole carbon source, consistent with the gene expression profile obtained from transcriptome sequencing.

Collectively, combined RT-qPCR and transcriptome results indicate that *P. kudriavzevii* P22 drives methanol conversion and utilization by upregulating the expression of key genes such as DAS, DAK, and FBP during methanol metabolism. The silencing of the AOX gene may imply that this strain utilizes alternative enzymes to convert methanol to formaldehyde.

Based on the above analysis, the methanol metabolism pathway in *P. kudriavzevii* P22 can be preliminarily deduced (Fig. 4e). After entering the cell, methanol is presumably dehydrogenated to formaldehyde via unannotated enzymes. Subsequently, formaldehyde is converted to dihydroxyacetone (DHA) through the dihydroxyacetone synthase (DAS)-mediated reaction. DHA is then phosphorylated by dihydroxyacetone kinase (DAK) to generate dihydroxyacetone phosphate (DHAP). Finally, DHAP is converted to fructose-6-phosphate (F6P) via the fructose-1,6-bisphosphatase (FBP)-catalyzed reaction, which further feeds into central carbon metabolism, thereby completing the assimilation of methanol.

Pichia kudriavzevii P22 produces single-cell protein through open-unsterilized fermentation process

In industrial biomanufacturing, the cost and conversion efficiency of carbon sources are key determinants of the production cost of high-value-added products. Given that the carbon source utilization spectrum of the screened *P. kudriavzevii* P22 remains unclear, we investigated its utilization of various carbon sources. As shown in Fig. 5a, at 24 h of fermentation, *P. kudriavzevii* P22 utilized glucose and fructose almost identically, with an OD_{600} value of 3.7–4.3-fold that of sucrose, 5.4-fold that of glycerol, and 6.9-fold that of raffinose. At 48 h of fermentation, cultures with glucose ($OD_{600} = 4.1$) and fructose ($OD_{600} = 4.4$) as carbon sources reached the stationary phase. In contrast, cultures with glycerol as the carbon source entered the exponential growth phase after an adaptation period, with the OD_{600} value reaching 3.9. By comparison, fermentation with sucrose and raffinose proceeded relatively slowly, with OD_{600} values of 1.5 and 0.9, respectively. After 48 h, glycerol-fed cultures remained in the exponential growth phase, entered the stationary phase at 96 h, and achieved a maximum OD_{600} of 6.3 at 120 h. In summary, *P. kudriavzevii* P22 exhibited robust utilization capacity for glycerol, fructose, and glucose, but low efficiency for carbon sources such as raffinose and sucrose. It could barely utilize starch, xylose, CMC-Na, or xylan. Thus, this strain is capable of utilizing not only methanol as a sole carbon source but also mixed carbon sources of methanol and other substrates, suggesting its potential for application in subsequent fermentation processes by selecting appropriate carbon sources based on practical needs.

pH and temperature are critical factors influencing microbial growth; accordingly, we optimized the optimal growth

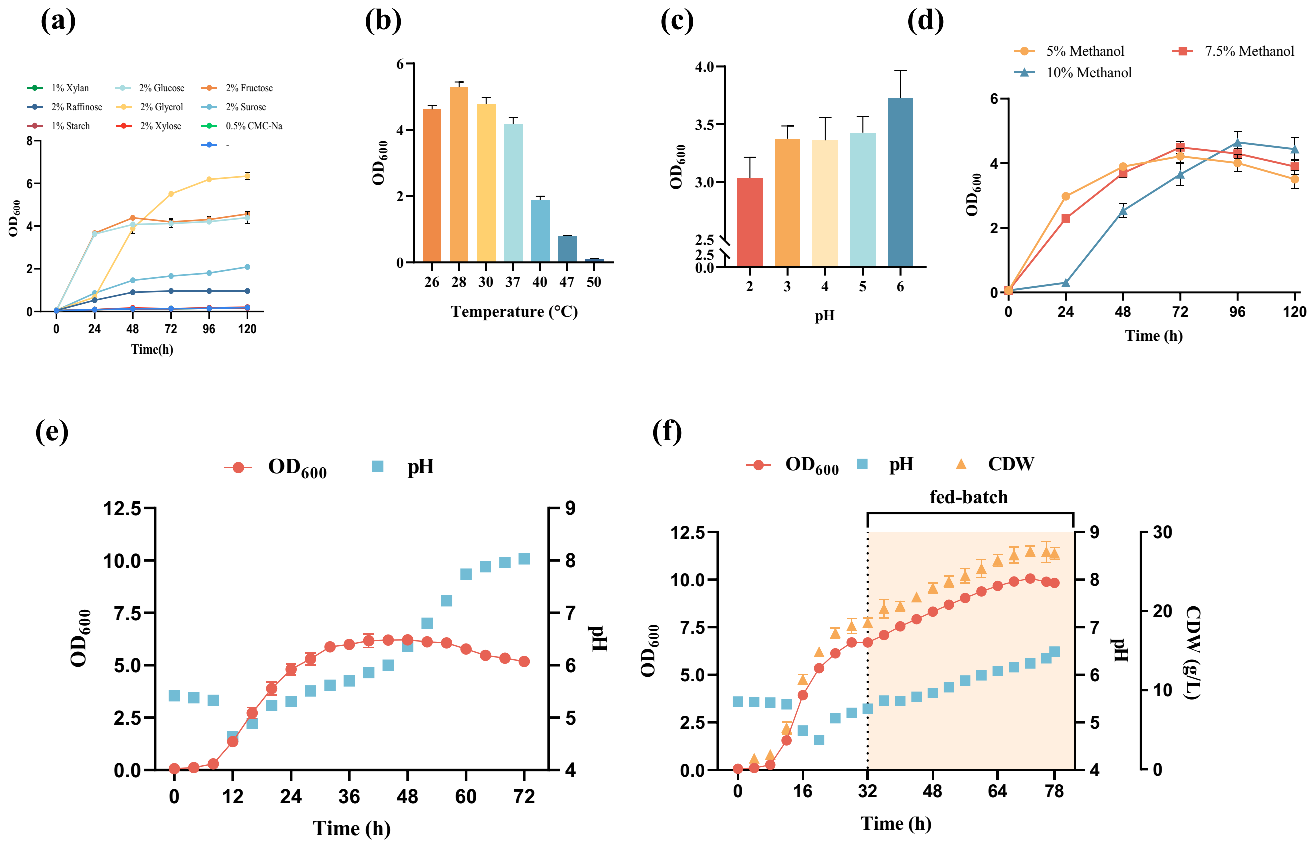

Fig. 5 **a** Utilization of *P. kudriavzevii* P22 carbon source. **b** *P. kudriavzevii* P22 is acid-resistant. **c** *P. kudriavzevii* P22's tolerance to high temperatures. **d** Growth status of *P. kudriavzevii* P22 in media with different methanol concentration gradients. **e** One-pot open-unsterilized fermentation OD600 with 7.5% (v/v) methanol added and pH. **f** Open feed fermentation OD600, pH, and CDW with 7.5% (v/v) methanol added

temperature and pH for *P. kudriavzevii* P22 fermentation. As shown in Fig. 5b, *P. kudriavzevii* P22 grew well within the temperature range of 26–37 °C, with OD_{600} values all exceeding 4, and the optimum growth temperature was 28 °C. However, with further temperature increases, the OD_{600} value decreased significantly, indicating high temperature sensitivity. As shown in Fig. 5c, the optimum pH for *P. kudriavzevii* P22 was 6 ($OD_{600}=3.7$), with similar growth performance within the pH range of 3–5 ($OD_{600}\approx3.4$). Notably, *P. kudriavzevii* P22 displayed high tolerance to low pH, as it could grow even at pH = 2 ($OD_{600}=3$), exhibiting greater acid tolerance than most yeast strains. Collectively, this suggests potential suitability for the fermentative production of acidic high-value-added products such as organic acids.

Methanol is toxic to most microorganisms. However, for *P. kudriavzevii* P22—capable of tolerating methanol and even utilizing it as the sole carbon source—methanol addition supports the feasibility of open-unsterilized fermentation. To explore its potential for single-cell protein production under open conditions, this study performed open-unsterilized fermentation shake flask experiments (Fig. 5 d) in media containing 5% (v/v), 7.5% (v/v), and 10% (v/v) methanol. As shown, at 24 h, the OD_{600} of cultures with 5% (v/v) methanol was 2.3, while that with 7.5% (v/v) methanol was 3.0; in contrast, the 10% (v/v) methanol system showed no significant growth ($OD_{600}=0.3$). By 72 h, the OD_{600} of the 7.5% (v/v) methanol culture (4.5) exceeded that of the 5% (v/v) culture (4.2), after which both entered the decline phase. The 10% (v/v) methanol system-initiated growth after 24 h, reached a maximum OD_{600} of 4.6 at 96 h with no bacterial contamination (Fig. SF2, 3), and then entered the decline phase. These results indicate that *P. kudriavzevii* P22 exhibits high methanol tolerance and utilization efficiency during fermentation, which is critical for contamination prevention in open-unsterilized fermentation.

Subsequently, the fermentation process was scaled up to a 5-L open system under 7.5% (v/v) methanol. Figure 5e shows the 72-h growth curve: *P. kudriavzevii* P22 remained in the adaptation phase during the first 8 h, entered the exponential growth phase from 8 to 32 h, maintained the stationary phase from 32 to 48 h, achieved maximum biomass

(OD_{600} = 6.2) at 44 h, and entered the decline phase after 48 h. Notably, yeast typically secretes acidic metabolites during fermentation, leading to gradual environmental acidification and continuous pH reduction. However, this pattern was not observed for *P. kudriavzevii* P22: in the first 8 h of one-pot fermentation (i.e., the adaptation phase), the broth pH initially decreased from 5.42 to ~ 5.33, then dropped rapidly to 4.64 at 12 h, and subsequently increased gradually to ~ 8 with prolonged fermentation. The variation in pH value observed during fermentation may be attributed to the reduced secretion of acidic substances and the increased synthesis of alkaline substances by *P. kudriavzevii* P22 during its fermentation process. To further enhance productivity, a fed-batch fermentation experiment was further conducted (Fig. 5f). Methanol feeding was initiated at 32 h (OD_{600} = 6.7). To confirm glucose depletion before feeding, glucose content in the 32-h broth (without methanol addition) was measured. Comparison with HPLC results of glucose standards confirmed complete glucose exhaustion prior to methanol feeding. After feeding, *P. kudriavzevii* P22 continued growth using methanol as the sole carbon source, reaching a maximum OD_{600} of 10.1 at 72 h, with a corresponding maximum cell dry weight (CDW) of 27.54 g/L. In fed-batch fermentation, pH decreased slowly from 5.44 to 5.38 within the first 12 h, then dropped rapidly to 4.63 at 20 h, increased sharply to 5.09 over the next 4 h, and rose gradually thereafter, reaching a maximum of 6.49 at 78 h.

Discussion

In this study, a strain of *P. kudriavzevii* P22 was isolated from fermented camel milk, harboring high robustness, tolerance to acidic condition (pH 2), high temperatures (47 °C), elevated methanol concentrations, and the ability to use methanol as the sole carbon source. In industrial fermentation, yeast cells commonly release acidic metabolites into the culture medium, gradually leading to environmental acidification. Acetic acid, a major by-product of microbial fermentation, is known to markedly inhibit microbial growth (Guaragnella and Bettiga 2021). Notably, the P22 strain was able to grow under 5 g/L acetic acid stress, outperforming *S. cerevisiae* S288C and *P. pastoris* GS115. Furthermore, microbial fermentation is exothermic, which elevates the temperature of the culture system and compromises yeast cell growth. Therefore, large volumes of cooling water are typically required for temperature control. Additionally, biomass fermentation generally requires enzymatic saccharification pretreatment at elevated temperatures (around 50 °C) before fermentation (Wang et al. 2018). Thermotolerant yeasts enable simultaneous saccharification and fermentation, thereby shortening fermentation duration, reducing cooling costs, and lowering contamination risks (Shahsavarani et al. 2013). Thus, thermotolerance is a critical trait for yeast-based fermentation processes. Consistent with this, previous studies have shown that *P. kudriavzevii* exhibits higher fermentation efficiency than *S. cerevisiae* at 35–45 °C. For instance, *P. kudriavzevii* RZ8-1 can achieve an ethanol yield of 3.38% (from 8.5% glucose) at 40 °C, and 7.86% for *P. kudriavzevii* DMKU 3-ET15 (Champnipa et al. 2018).

During fermentation, culture media often contain high concentrations of essential salt ions, resulting in increased osmotic pressure within the system. Such conditions can inhibit normal microbial proliferation, even cause cell death, ultimately reducing final product yields. Moreover, high salt concentrations help suppress contamination by heterotrophic bacteria, thereby enabling non-sterilized open-unsterilized fermentation systems and lowering contamination control costs. *P. kudriavzevii* P22 can survive in hyperosmotic environments (100 g/L NaCl), exhibiting promising potential in fermentation systems with high osmotic pressure.

Methanol, as an economical and sustainable carbon source, provides low feedstock costs, established production processes, and high compatibility with existing industrial infrastructure. However, methanol is toxic to most microorganisms; methanol itself primarily damages cell membranes, while its metabolite formaldehyde damages proteins. The responses to methanol of yeasts seem to be strain-specific, associated with mitochondrial metabolism, and peroxisomal function (Joan Lin Cereghino 2000). Methylotrophic microorganisms assimilate methanol efficiently through specialized metabolic pathways, using it as the sole carbon and energy source to synthesize high-value-added products. Their industrial potential has been demonstrated in the synthesis of polymers (e.g., polyhydroxybutyrate, PHB), and their ability to accumulate natural products supports the development of a methanol-based bioeconomy (Wang et al. 2023). *P. kudriavzevii* P22 can tolerate methanol concentrations up to 15%, the highest reported for wild-type yeasts to date, showing potential for industrial application under laboratory-scale conditions. Furthermore, studies have indicated that phenyl-ethanol—a metabolite of *P. kudriavzevii*—significantly inhibits the growth of competing fungi, which may further promote its application in open fermentation (van Rijswijck et al. 2014).

Compared with closed systems that require sophisticated environmental controls, substantial investment, and complex operation, open systems offer notable advantages. Open fermentation systems reduce costs through simpler design, enhanced operational flexibility, improved production scale scalability, and mitigated cost for contamination. Consequently, researchers have developed open non-sterilized fermentation processes using bacterial chassis (Zhang et al. 2017). Due to its stress tolerance, the *P. kudriavzevii* P22 strain isolated in this study shows promise for open non-sterilized fermentation and provides a valuable resource for advancing yeast-based open non-sterilized fermentation technologies.

Currently, approximately 65% of regions worldwide lack regulatory frameworks for single-cell protein (SCP), limiting industrialization due to insufficient food safety standards and gaps in allergen risk assessment. *P. kudriavzevii* is widely distributed in nature (Bourdichon et al. 2012; del Mónaco et al. 2014), classified as Biosafety Level 1 (BSL-1), and designated as safe for consumption by the U.S. Food and Drug Administration (FDA) (Mitchell et al. 2018). Besides, *P. kudriavzevii* P22 in this study is non-genetically engineered. Thus, open SCP production with *P. kudriavzevii* P22 not only circumvents ethical issues associated with regulatory gaps but also reduces production costs through process simplification, offering a flexible and practical approach for SCP production and an economical, safe solution for C1 resource bioconversion. Nevertheless, open-unsterilized fermentation of *P. kudriavzevii* P22 has certain limitations. First, compared with high-density fermentation strains, P22 achieves lower biomass yield, with a notable performance gap (Abeln and Chuck 2019). Second, as a non-model strain, P22 lacks comprehensive genetic characterization and molecular tools, suffers from incomplete genome annotation, and has limited mechanistic research. Additionally, although *P. kudriavzevii* P22 exhibits greater methanol tolerance than common methylotrophic yeasts (e.g., *P. pastoris* GS115), future efforts are required to address bottlenecks in strain adaptability and large-scale production costs.

This study highlights the potential of *P. kudriavzevii* P22, a methylotrophic yeast isolated from fermented camel milk, as a robust chassis for open fermentation-based SCP production. Its exceptional tolerance to methanol, acid, temperature, and osmotic stress provides a solid foundation for developing non-sterile, low-cost bioprocesses. Future research should focus on elucidating its methanol metabolism and stress response networks, scaling up dynamic process control, and leveraging its synthetic biology potential through compartmentalized pathways and genome engineering. Moreover, integrating SCP production with local agricultural systems offers opportunities for establishing closed-loop, carbon-reducing industrial chains. Furthermore, this strategy provides an opportunity to restructure the feed protein industry. In summary, *P. kudriavzevii* P22 constitutes a promising laboratory-scale platform for methanol-based biomanufacturing, with the potential to advance the sustainability and resilience of the global protein supply chain. Although the present work demonstrates its robustness and feasibility at the 5-L scale, further validation under pilot- or continuous-operation conditions is warranted. As a non-model yeast, *P. kudriavzevii* P22 remains insufficiently characterized at the genomic and molecular levels, and its long-term stability during extended or continuous fermentation requires additional investigation.

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Data availability Data is provided within the manuscript or supplementary information files. Due to the capacity limitations, a portion of the raw sequencing data is available in Mendeley Data (https://doi.org/10.17632/ffkmpvpcm9.3). A complete version can be obtained on request.

Declarations

Ethical approval Not applicable.

Competing interests The authors declare no competing interests.

References

Abeln F, Chuck CJ (2019) Achieving a high-density oleaginous yeast culture: comparison of four processing strategies using *Metschnikowia pulcherrima*. Biotechnol Bioeng 116(12):3200–3214. https://doi.org/10.1002/bit.27141

Bajić B, Vučurović D, Vasić Đ, Jevtić-Mučibabić R, Dodić S (2022) Biotechnological production of sustainable microbial proteins from agro-industrial residues and by-products. Foods. https://doi.org/10.3390/foods12010107

Bourdichon F, Casaregola S, Farrokh C, Frisvad JC, Gerds ML, Hammes WP, Harnett J, Huys G, Laulund S, Ouwehand A, Powell IB, Prajapati JB, Seto Y, Ter Schure E, Van Boven A, Vankerckhoven V, Zgoda A, Tuijtelaars S, Hansen EB (2012) Erratum to

"Food fermentations: microorganisms with technological beneficial use" [International Journal of Food Microbiology 154 (2012) 87–97]. Int J Food Microbiol. https://doi.org/10.1016/j.ijfoodmicro.2012.04.003

Cai P, Wu X, Deng J, Gao L, Shen Y, Yao L, Zhou YJ (2022) Methanol biotransformation toward high-level production of fatty acid derivatives by engineering the industrial yeast *Pichia pastoris*. Proc Natl Acad Sci U S A 119(29):e2201711119. https://doi.org/10.1073/pnas.2201711119

Chamnipa N, Thanonkeo S, Klanrit P, Thanonkeo P (2018) The potential of the newly isolated thermotolerant yeast *Pichia kudriavzevii* RZ8-1 for high-temperature ethanol production. Braz J Microbiol 49(2):378–391. https://doi.org/10.1016/j.bjm.2017.09.002

Cotton CAR, Claassens NJ, Benito-Vaquerizo S, Bar-Even A (2020) Renewable methanol and formate as microbial feedstocks. Curr Opin Biotechnol 62:168–180. https://doi.org/10.1016/j.copbio.2019.10.002

del Mónaco SM, Barda NB, Rubio NC, Caballero AC (2014) Selection and characterization of a Patagonian *Pichia kudriavzevii* for wine deacidification. J Appl Microbiol 117(2):451–464. https://doi.org/10.1111/jam.12547

Fabarius JT, Wegat V, Roth A, Sieber V (2021) Synthetic methylotrophy in yeasts: towards a circular bioeconomy. Trends Biotechnol 39(4):348–358. https://doi.org/10.1016/j.tibtech.2020.08.008

Gao L, Khoo SC, Zhang Z, Wu X (2025) Trends in sustainable single-cell protein from non-grain feedstocks. Trends Biotechnol. https://doi.org/10.1016/j.tibtech.2025.04.018

Guaragnella N, Bettiga M (2021) Acetic acid stress in budding yeast: from molecular mechanisms to applications. Yeast 38(7):391–400. https://doi.org/10.1002/yea.3651

Guo F, Qiao Y, Xin F, Zhang W, Jiang M (2023) Bioconversion of C1 feedstocks for chemical production using *Pichia pastoris*. Trends Biotechnol 41(8):1066–1079. https://doi.org/10.1016/j.tibtech.2023.03.006

Jia M, Shao L, Jiang J, Jiang W, Xin F, Zhang W, Jiang Y, Jiang M (2024) Mitigating toxic formaldehyde to promote efficient utilization of C1 resources. Crit Rev Biotechnol 45(5):1175–1187. https://doi.org/10.1080/07388551.2024.2430476

Joan Lin Cereghino JMC (2000) Heterologous protein expression in the methylotrophic yeast *Pichia pastoris*. doi.org/https://doi.org/10.1111/j.1574-6976.2000.tb00532.x.

Kelso PA, Chow LKM, Carpenter AC, Paulsen IT, Williams TC (2022) Toward Methanol-Based Biomanufacturing: Emerging Strategies for Engineering Synthetic Methylotrophy in Saccharomyces cerevisiae. ACS Synth Biol 11(8):2548–2563. https://doi.org/10.1021/acssynbio.2c00110

Li YP, Ahmadi F, Kariman K, Lackner M (2024) Recent advances and challenges in single cell protein (SCP) technologies for food and feed production. npj Science of Food 8(1) https://doi.org/10.1038/s41538-024-00299-2

Meng J, Liu S, Gao L, Hong K, Liu S, Wu X (2023) Economical production of *Pichia pastoris* single cell protein from methanol at industrial pilot scale. Microb Cell Fact. https://doi.org/10.1186/s12934-023-02198-9

Mitchell AP, Douglass AP, Offei B, Braun-Galleani S, Coughlan AY, Martos AAR, Ortiz-Merino RA, Byrne KP, Wolfe KH (2018) Population genomics shows no distinction between pathogenic Candida krusei and environmental *Pichia kudriavzevii:* One species, four names. PLOS Pathogens 14(7) https://doi.org/10.1371/journal.ppat.1007138

Sá AGA, Moreno YMF, Carciofi BAM (2020) Plant proteins as high-quality nutritional source for human diet. Trends Food Sci Technol 97:170–184. https://doi.org/10.1016/j.tifs.2020.01.011

Sakarika M, Ganigué R, Rabaey K (2022) Methylotrophs: from C1 compounds to food. Current Opinion in Biotechnology 75 https://doi.org/10.1016/j.copbio.2022.102685

Salazar-López NJ, Barco-Mendoza GA, Zuñiga-Martínez BS, Domínguez-Avila JA, Robles-Sánchez RM, Ochoa MAV, González-Aguilar GA (2022) Single-cell protein production as a strategy to reincorporate food waste and agro by-products back into the processing chain. Bioengineering. https://doi.org/10.3390/bioengineering9110623

Sekoai PT, Roets-Dlamini Y, O'Brien F, Ramchuran S, Chunilall V (2024) Valorization of food waste into single-cell protein: an innovative technological strategy for sustainable protein production. Microorganisms. https://doi.org/10.3390/microorganisms12010166

Shahsavarani H, Hasegawa D, Yokota D, Sugiyama M, Kaneko Y, Boonchird C, Harashima S (2013) Enhanced bio-ethanol production from cellulosic materials by semi-simultaneous saccharification and fermentation using high temperature resistant Saccharomyces cerevisiae TJ14. J Biosci Bioeng 115(1):20–23. https://doi.org/10.1016/j.jbiosc.2012.07.018

Smith K, Watson AW, Lonnie M, Peeters WM, Oonincx D, Tsoutsoura N, Simon-Miquel G, Szepe K, Cochetel N, Pearson AG, Witard OC, Salter AM, Bennett M, Corfe BM (2024) Meeting the global protein supply requirements of a growing and ageing population. Eur J Nutr 63(5):1425–1433. https://doi.org/10.1007/s00394-024-03358-2

Steensels J, Snoek T, Meersman E, Nicolino MP, Voordeckers K, Verstrepen KJ (2014) Improving industrial yeast strains: exploiting natural and artificial diversity. FEMS Microbiol Rev 38(5):947–995. https://doi.org/10.1111/1574-6976.12073

Su T, Le B, Zhang W, Bak KH, Soladoye PO, Zhao Z, Zhao Y, Fu Y, Wu W (2024) Corrigendum to "Technological challenges and future perspectives of plant-based meat analogues: from the viewpoint of proteins" [Food Res. Int. 186 (2024) 114351]. Food Res Int. https://doi.org/10.1016/j.foodres.2024.114472

van Rijswijck IMH, Dijksterhuis J, Wolkers-Rooijackers JCM, Abee T, Smid EJ (2014) Nutrient limitation leads to penetrative growth into agar and affects aroma formation in P*ichia fabianii*. *Pichia Kudriavzevii* and Saccharomyces Cerevisiae yeast:n/a-n/a. https://doi.org/10.1002/yea.3050

Vasudevan UM, Soussan L, Fernández-Castané A, Noack S, Goddard AD (2025) Yeast as C1 cell factory: Transforming methanol and Formate into high-value compounds. Biotechnology Advances 83 https://doi.org/10.1016/j.biotechadv.2025.108641

Vethathirri RS, Santillan E, Wuertz S (2021) Microbial community-based protein production from wastewater for animal feed applications. Bioresource Technology 341 https://doi.org/10.1016/j.biortech.2021.125723

Wang Y, Cai D, He M, Wang Z, Qin P, Tan T (2015) Open fermentative production of l-lactic acid using white rice bran by simultaneous saccharification and fermentation. Bioresour Technol 198:664–672. https://doi.org/10.1016/j.biortech.2015.09.010

Wang L, Zhang K, Xu Y, Zhang M, Wang D (2018) High-solid pretreatment of corn stover using urea for enzymatic saccharification. Bioresour Technol 259:83–90. https://doi.org/10.1016/j.biortech.2018.03.023

Wang J, Chen Z, Deng X, Yuan Q, Ma H (2023) Engineering *Escherichia coli* for Poly-β-hydroxybutyrate production from methanol. Bioengineering. https://doi.org/10.3390/bioengineering10040415

Wegat V, Fabarius JT, Sieber V (2022) Synthetic methylotrophic yeasts for the sustainable fuel and chemical production. Biotechnology for Biofuels and Bioproducts 15(1) https://doi.org/10.1186/s13068-022-02210-1

Ye J-W, Lin Y-N, Yi X-Q, Yu Z-X, Liu X, Chen G-Q (2023) Synthetic biology of extremophiles: a new wave of biomanufacturing. Trends Biotechnol 41(3):342–357. https://doi.org/10.1016/j.tibtech.2022.11.010

Ye L, Bogicevic B, Bolten CJ, Wittmann C (2024) Single-cell protein: overcoming technological and biological challenges towards improved industrialization. Current Opinion in Biotechnology 88 https://doi.org/10.1016/j.copbio.2024.103171

Yue H, Ling C, Yang T, Chen X, Chen Y, Deng H, Wu Q, Chen J, Chen G-Q (2014) A seawater-based open and continuous process for polyhydroxyalkanoates production by recombinant *Halomonas campaniensis* LS21 grown in mixed substrates. Biotechnol Biofuels 7(1):108. https://doi.org/10.1186/1754-6834-7-108

Zang X, Yang Y, Zhan C, Bai Z (2025) Methanol metabolism in synthetic methylotrophic microorganisms. Biotechnology Advances 83 https://doi.org/10.1016/j.biotechadv.2025.108623

Zhang C, Zhou C, Assavasirijinda N, Yu B, Wang L, Ma Y (2017) Non-sterilized fermentation of high optically pure d-lactic acid by a genetically modified thermophilic Bacillus coagulans strain. Microbial Cell Factories 16(1) https://doi.org/10.1186/s12934-017-0827-1

Zhu T, Zhao T, Bankefa OE, Li Y (2020) Engineering unnatural methylotrophic cell factories for methanol-based biomanufacturing: Challenges and opportunities. Biotechnology Advances 39 https://doi.org/10.1016/j.biotechadv.2019.107467